

Design Data for CAD model and drawing preparation

Human-readable handoff of the saved approved-component geometry. Use the dimensions and coordinate schedules, not scaled illustrations. This report does not regenerate, resize or qualify the equipment.

Document authority	Saved record
Design / revision / record	269 / 3 / 3
Saved on	2026-09-20T12:20:11.436Z
Currentness	CURRENT
Geometry status	SYSTEM-GENERATED APPROVED TURBINE/PERFORATED-STATOR PRE-PILOT GEOMETRY — GEOMETRICALLY COMPLETE
Geometry ruleset	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR
Source Stage 3 ID	64
Source Stage 4 ID	15
Combined source SHA-256	04dfb56f2139741bb5813460c9a575e95b6927718f837ed09629fb70e2bbb610
Geometry SHA-256	d99289ce7f5b194f460b8cb75ac1396619b51bf5af2d84227e3d781979eadae6
Rules manifest SHA-256	02f4ea7a17ce607c51df2e90c56f0ed7c689b92a0714f40cdf63db54abc7047

A = frozen upstream authority; B = documented reference feature; C = approved R1 engineering envelope. A calculated Class-C value remains Class C. Geometry completeness does not establish hydraulic, structural or fabrication adequacy.

Approved frozen shrouded turbine and 84-hole perforated stator integrated; inherited 18-compartment / 3240-mm active stack. Historical revisions preserved. Preliminary, not for fabrication.

Coordinate and datum convention

Right-handed Cartesian coordinates. Origin (0,0,0) is the bottom vessel pole on the shaft/column axis. +Z is vertically upward; X/Y are the horizontal plane. Azimuth 0° is +X, 90° is +Y, increasing counter-clockwise when viewed from above (+Z toward origin).

All length tables and XYZ coordinates below are millimetres; angular values are degrees. The immutable source stores SI metres. Negative Z values belong to the skirt below the bottom vessel pole. Radial shell connections point outward; “up” and “down” connections have vertical axes. Connection centre denotes saved vessel-surface intersection, and end denotes saved external stub termination, not a flange face.

Numeric presentation is rounded to at most six decimal places, without modifying source values or adding manufacturing tolerances. Symbols in the provenance rules refer to SI source quantities: D = column ID, DR = rotor swept diameter, hc = compartment pitch, ds = shaft diameter, do = opening diameter, ts = stator thickness, HR = rotor axial envelope, dh = hub diameter, tb = blade thickness, Lb = blade radial length, HA = installed active height. Do not mix mm table values with SI rule quantities.

Rotor vertex XYZ values are local to the rotor centre plane (Z=0) on the shared shaft axis. Translate local Z by the rotor elevation in the complete schedule; do not apply additional clocking. Support footprints are global XY; combine them with the saved arm bottom/top Z values. Head profiles are internal envelopes, not material outer surfaces.

Design basis and source provenance

Design feature	Saved value	Unit	Provenance / governing rule
Stage3 Result Id	64	—	Class A; Inherited ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Stage4 Result Id	15	—	Class A; Inherited ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Column Diameter	600	mm	Class A; Inherited ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Compartment Height	180	mm	Class A; Inherited ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Rotor Diameter Ratio	0.33	—	Class A; Inherited ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Rotor Diameter	198	mm	Class A; Inherited ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Selected Rpm	45	rpm	Class A; Inherited ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Rpm Min	30	rpm	Class A; Inherited ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Rpm Max	60	rpm	Class A; Inherited ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Phase Configuration	nmp-continuous- rrbo-dispersed	—	Class A; Inherited ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Compartment Count	18	—	Class A; Inherited ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Required Active Height	3240	mm	Class A; Inherited ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Installed Active Height	3240	mm	Class A; Inherited ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.

Design feature	Saved value	Unit	Provenance / governing rule
Design Nt	7	—	Class A; Inherited ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Column Area area	282743.338823	mm ²	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. $\pi D^2/4$.
Compartment1-bottom	750	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment1-top	930	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment1-rotor	840	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment2-bottom	930	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment2-top	1110	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment2-rotor	1020	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment3-bottom	1110	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment3-top	1290	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment3-rotor	1200	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment4-bottom	1290	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment4-top	1470	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment4-rotor	1380	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

Design feature	Saved value	Unit	Provenance / governing rule
Compartment5-bottom	1470	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment5-top	1650	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment5-rotor	1560	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment6-bottom	1650	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment6-top	1830	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment6-rotor	1740	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment7-bottom	1830	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment7-top	2010	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment7-rotor	1920	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment8-bottom	2010	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment8-top	2190	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment8-rotor	2100	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment9-bottom	2190	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment9-top	2370	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

Design feature	Saved value	Unit	Provenance / governing rule
Compartment9-rotor	2280	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment10-bottom	2370	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment10-top	2550	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment10-rotor	2460	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment11-bottom	2550	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment11-top	2730	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment11-rotor	2640	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment12-bottom	2730	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment12-top	2910	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment12-rotor	2820	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment13-bottom	2910	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment13-top	3090	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment13-rotor	3000	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment14-bottom	3090	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

Design feature	Saved value	Unit	Provenance / governing rule
Compartment14-top	3270	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment14-rotor	3180	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment15-bottom	3270	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment15-top	3450	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment15-rotor	3360	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment16-bottom	3450	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment16-top	3630	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment16-rotor	3540	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment17-bottom	3630	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment17-top	3810	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment17-rotor	3720	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment18-bottom	3810	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment18-top	3990	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.
Compartment18-rotor	3900	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Frozen pitch/count plus R1 datum; no independent drawing recalculation.

Vessel, head and active/end-zone geometry

Design feature	Saved value	Unit	Provenance / governing rule
Bottom Disengagement	600	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D; packaging allocation, not separation-time qualification.
Top Disengagement	600	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D; packaging allocation, not separation-time qualification.
Bottom Head Depth	150	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D/4; 2:1 ellipsoidal internal envelope.
Top Head Depth	150	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D/4; 2:1 ellipsoidal internal envelope.
Head Straight Flange	0	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Zero in R1 internal envelope model.
Active Start	750	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. hB+LB.
Active End	3990	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. z0+HA.
Top Tangent	4590	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. z1+LT.
Vessel Height	4740	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. HA+2.5D.
Overall Height	5640	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Support+vessel+drive = HA+4D.
Head Profiles	2:1 ellipsoidal	—	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Internal envelope only.

Head	Tangent Z mm	Pole Z mm	Radial semiaxis mm	Axial semiaxis mm
bottom	150	0	300	150
top	4590	4740	300	150

Each head is the saved half-ellipsoid of revolution: $r^2/a^2 + (z - \text{tangent Z})^2/b^2 = 1$, with semiaxes a/b listed above. Select the half between tangent and pole. This describes the internal geometric envelope only.

The vessel straight internal cylinder uses the frozen column ID and extends between the bottom and top head tangent elevations. No shell outside diameter or material wall thickness is inferred. Model reserved envelopes separately from material-bearing components.

Rotor geometry — approved double-entry shrouded six-blade assembly

Approved double-entry radial-flow shrouded six-blade turbine. The saved solids below are the exact approved preliminary component definition; this is not fabrication qualification.

Design feature	Saved value	Unit	Provenance / governing rule
Rotor Offset	90	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. hc/2; centred rotor.
Blade Azimuths angle	0, 60, 120, 180, 240, 300	deg	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Six equally spaced blades; same clocking in every compartment.
Blade Count	6	—	Class C; System-generated Class C. Approved six-blade architecture.
Blade Height	28	mm	Class C; System-generated Class C. Approved outer paddle height.
Blade Inner Segment Height	16	mm	Class C; System-generated Class C. Approved inner web height.
Blade Outer Segment Height	28	mm	Class C; System-generated Class C. Approved outer paddle height.
Blade Thickness	3	mm	Class C; System-generated Class C. Approved blade thickness.
Hub Diameter	70	mm	Class C; System-generated Class C. Approved hub diameter.
Hub Height	24	mm	Class C; System-generated Class C. Approved hub height.
Rotor Axial Envelope	32	mm	Class C; System-generated Class C. Approved total shrouded rotor height.
Blade Radial Length	64	mm	Class C; System-generated Class C. Approved nominal radial reach from hub OD to rotor OD.
Blade Inner Segment Length	30	mm	Class C; System-generated Class C. Approved web radial interval, 35 to 65 mm radius.
Blade Outer Segment Length	34	mm	Class C; System-generated Class C. Approved paddle/shroud radial interval, 65 to 99 mm radius.
Shroud Thickness	2	mm	Class C; System-generated Class C. Each approved upper/lower shroud.
Shroud Inner Diameter	130	mm	Class C; System-generated Class C. Approved eye diameter.
Shroud Outer Diameter	198	mm	Class C; System-generated Class C. Approved rotor/shroud OD.
Rotor Construction	approved-double-entry-shrouded-turbine	—	Class C; System-generated Class C. Approved component architecture identifier.

Blade number	Saved azimuth °
1	0

Blade number	Saved azimuth °
2	60
3	120
4	180
5	240
6	300

Profile vertex	Radial X mm	Local Z mm
1	35	-8
2	65	-8
3	65	-14
4	99	-14
5	99	14
6	65	14
7	65	8
8	35	8

Close the ordered polygon back to vertex 1. The following already-clocked saved 3D vertices define each blade envelope; first and second polygon loops are the opposing thickness faces. Connect corresponding loop vertices. No new blade dimensions are inferred.

Blade / azimuth	Vertex	Local X, Y, Z mm
1 / 0°	1	34.967842, -1.5, -8
1 / 0°	2	64.98269, -1.5, -8
1 / 0°	3	64.98269, -1.5, -14
1 / 0°	4	98.988636, -1.5, -14
1 / 0°	5	98.988636, -1.5, 14
1 / 0°	6	64.98269, -1.5, 14
1 / 0°	7	64.98269, -1.5, 8
1 / 0°	8	34.967842, -1.5, 8
1 / 0°	9	34.967842, 1.5, -8
1 / 0°	10	64.98269, 1.5, -8
1 / 0°	11	64.98269, 1.5, -14
1 / 0°	12	98.988636, 1.5, -14
1 / 0°	13	98.988636, 1.5, 14
1 / 0°	14	64.98269, 1.5, 14
1 / 0°	15	64.98269, 1.5, 8
1 / 0°	16	34.967842, 1.5, 8
2 / 60°	1	18.782959, 29.53304, -8

Blade / azimuth	Vertex	Local X, Y, Z mm
2 / 60°	2	33.790383, 55.52666, -8
2 / 60°	3	33.790383, 55.52666, -14
2 / 60°	4	50.793356, 84.976673, -14
2 / 60°	5	50.793356, 84.976673, 14
2 / 60°	6	33.790383, 55.52666, 14
2 / 60°	7	33.790383, 55.52666, 8
2 / 60°	8	18.782959, 29.53304, 8
2 / 60°	9	16.184883, 31.03304, -8
2 / 60°	10	31.192307, 57.02666, -8
2 / 60°	11	31.192307, 57.02666, -14
2 / 60°	12	48.19528, 86.476673, -14
2 / 60°	13	48.19528, 86.476673, 14
2 / 60°	14	31.192307, 57.02666, 14
2 / 60°	15	31.192307, 57.02666, 8
2 / 60°	16	16.184883, 31.03304, 8
3 / 120°	1	-16.184883, 31.03304, -8
3 / 120°	2	-31.192307, 57.02666, -8
3 / 120°	3	-31.192307, 57.02666, -14
3 / 120°	4	-48.19528, 86.476673, -14
3 / 120°	5	-48.19528, 86.476673, 14
3 / 120°	6	-31.192307, 57.02666, 14
3 / 120°	7	-31.192307, 57.02666, 8
3 / 120°	8	-16.184883, 31.03304, 8
3 / 120°	9	-18.782959, 29.53304, -8
3 / 120°	10	-33.790383, 55.52666, -8
3 / 120°	11	-33.790383, 55.52666, -14
3 / 120°	12	-50.793356, 84.976673, -14
3 / 120°	13	-50.793356, 84.976673, 14
3 / 120°	14	-33.790383, 55.52666, 14
3 / 120°	15	-33.790383, 55.52666, 8
3 / 120°	16	-18.782959, 29.53304, 8
4 / 180°	1	-34.967842, 1.5, -8
4 / 180°	2	-64.98269, 1.5, -8
4 / 180°	3	-64.98269, 1.5, -14
4 / 180°	4	-98.988636, 1.5, -14

Blade / azimuth	Vertex	Local X, Y, Z mm
4 / 180°	5	-98.988636, 1.5, 14
4 / 180°	6	-64.98269, 1.5, 14
4 / 180°	7	-64.98269, 1.5, 8
4 / 180°	8	-34.967842, 1.5, 8
4 / 180°	9	-34.967842, -1.5, -8
4 / 180°	10	-64.98269, -1.5, -8
4 / 180°	11	-64.98269, -1.5, -14
4 / 180°	12	-98.988636, -1.5, -14
4 / 180°	13	-98.988636, -1.5, 14
4 / 180°	14	-64.98269, -1.5, 14
4 / 180°	15	-64.98269, -1.5, 8
4 / 180°	16	-34.967842, -1.5, 8
5 / 240°	1	-18.782959, -29.53304, -8
5 / 240°	2	-33.790383, -55.52666, -8
5 / 240°	3	-33.790383, -55.52666, -14
5 / 240°	4	-50.793356, -84.976673, -14
5 / 240°	5	-50.793356, -84.976673, 14
5 / 240°	6	-33.790383, -55.52666, 14
5 / 240°	7	-33.790383, -55.52666, 8
5 / 240°	8	-18.782959, -29.53304, 8
5 / 240°	9	-16.184883, -31.03304, -8
5 / 240°	10	-31.192307, -57.02666, -8
5 / 240°	11	-31.192307, -57.02666, -14
5 / 240°	12	-48.19528, -86.476673, -14
5 / 240°	13	-48.19528, -86.476673, 14
5 / 240°	14	-31.192307, -57.02666, 14
5 / 240°	15	-31.192307, -57.02666, 8
5 / 240°	16	-16.184883, -31.03304, 8
6 / 300°	1	16.184883, -31.03304, -8
6 / 300°	2	31.192307, -57.02666, -8
6 / 300°	3	31.192307, -57.02666, -14
6 / 300°	4	48.19528, -86.476673, -14
6 / 300°	5	48.19528, -86.476673, 14
6 / 300°	6	31.192307, -57.02666, 14
6 / 300°	7	31.192307, -57.02666, 8

Blade / azimuth	Vertex	Local X, Y, Z mm
6 / 300°	8	16.184883, -31.03304, 8
6 / 300°	9	18.782959, -29.53304, -8
6 / 300°	10	33.790383, -55.52666, -8
6 / 300°	11	33.790383, -55.52666, -14
6 / 300°	12	50.793356, -84.976673, -14
6 / 300°	13	50.793356, -84.976673, 14
6 / 300°	14	33.790383, -55.52666, 14
6 / 300°	15	33.790383, -55.52666, 8
6 / 300°	16	18.782959, -29.53304, 8

Shroud	Inner / outer radius mm	Bottom / top local Z mm
lower	65 / 99	-16 / -14
upper	65 / 99	14 / 16

Approved eye diameter 130 mm; inner web height 16 mm; outer paddle height 28 mm. The upper and lower shrouds are separate annular solids.

Stator geometry and area convention

Approved physical convention: gross opening fraction = $(DSO^2 + N dh^2)/D^2 = 0.40$. The shaft-corrected value is diagnostic only and is not fed upstream.

Design feature	Saved value	Unit	Provenance / governing rule
Stator Free Area Ratio	0.4	—	Class A; Inherited ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Stator Gross Open Area area	113097.335529	mm ²	Class C; System-generated recomputed from approved component solids; no upstream feedback.
Stator Net Open Area area	111576.804685	mm ²	Class C; System-generated recomputed from approved component solids; no upstream feedback.
Gross Free Area Ratio	0.4	—	Class C; System-generated recomputed from approved component solids; no upstream feedback.
Shaft Blocked Free Area Ratio	0.394622	—	Class C; System-generated recomputed from approved component solids; no upstream feedback.
Calculated Free Area Ratio	0.4	—	Class C; System-generated recomputed from approved component solids; no upstream feedback.
Stator Radial Width	244	mm	Class C; System-generated recomputed from approved component solids; no upstream feedback.
Stator Thickness	4	mm	Class C; System-generated Class C. Approved plate thickness.
Stator Opening Diameter	112	mm	Class C; System-generated Class C. Approved central opening (130 - 2×9).
Stator Hole Count	84	—	Class C; System-generated Class C. Approved exact perforation count.
Stator Hole Diameter	39.559479	mm	Class C; System-generated Class C. Approved calculated profile-hole diameter.
Stator Lane Width	8	mm	Class C; System-generated Class C. Six retained radial no-hole lanes.
Stator Construction	approved-perforated-stator	—	Class C; System-generated Class C. Approved perforated plate architecture identifier.

Row	PCD mm	Count	First angle °	Step °
1	200	12	15	30
2	310	18	10	20
3	420	24	7.5	15
4	530	30	6	12

Row / hole	Exact X mm	Exact Y mm	Angle °
1 / 1	96.592583	25.881905	15
1 / 2	70.710678	70.710678	45

Row / hole	Exact X mm	Exact Y mm	Angle °
1 / 3	25.881905	96.592583	75
1 / 4	-25.881905	96.592583	105
1 / 5	-70.710678	70.710678	135
1 / 6	-96.592583	25.881905	165
1 / 7	-96.592583	-25.881905	195
1 / 8	-70.710678	-70.710678	225
1 / 9	-25.881905	-96.592583	255
1 / 10	25.881905	-96.592583	285
1 / 11	70.710678	-70.710678	315
1 / 12	96.592583	-25.881905	345
2 / 1	152.645202	26.915468	10
2 / 2	134.233938	77.5	30
2 / 3	99.63208	118.736889	50
2 / 4	53.013122	145.652356	70
2 / 5	0	155	90
2 / 6	-53.013122	145.652356	110
2 / 7	-99.63208	118.736889	130
2 / 8	-134.233938	77.5	150
2 / 9	-152.645202	26.915468	170
2 / 10	-152.645202	-26.915468	190
2 / 11	-134.233938	-77.5	210
2 / 12	-99.63208	-118.736889	230
2 / 13	-53.013122	-145.652356	250
2 / 14	0	-155	270
2 / 15	53.013122	-145.652356	290
2 / 16	99.63208	-118.736889	310
2 / 17	134.233938	-77.5	330
2 / 18	152.645202	-26.915468	350
3 / 1	208.203421	27.4105	7.5
3 / 2	194.014702	80.363521	22.5
3 / 3	166.604201	127.8399	37.5
3 / 4	127.8399	166.604201	52.5
3 / 5	80.363521	194.014702	67.5
3 / 6	27.4105	208.203421	82.5
3 / 7	-27.4105	208.203421	97.5

Row / hole	Exact X mm	Exact Y mm	Angle °
3 / 8	-80.363521	194.014702	112.5
3 / 9	-127.8399	166.604201	127.5
3 / 10	-166.604201	127.8399	142.5
3 / 11	-194.014702	80.363521	157.5
3 / 12	-208.203421	27.4105	172.5
3 / 13	-208.203421	-27.4105	187.5
3 / 14	-194.014702	-80.363521	202.5
3 / 15	-166.604201	-127.8399	217.5
3 / 16	-127.8399	-166.604201	232.5
3 / 17	-80.363521	-194.014702	247.5
3 / 18	-27.4105	-208.203421	262.5
3 / 19	27.4105	-208.203421	277.5
3 / 20	80.363521	-194.014702	292.5
3 / 21	127.8399	-166.604201	307.5
3 / 22	166.604201	-127.8399	322.5
3 / 23	194.014702	-80.363521	337.5
3 / 24	208.203421	-27.4105	352.5
4 / 1	263.548302	27.700043	6
4 / 2	252.029977	81.889504	18
4 / 3	229.496732	132.5	30
4 / 4	196.933379	177.319611	42
4 / 5	155.763092	214.389504	54
4 / 6	107.78521	242.089546	66
4 / 7	55.096598	259.209114	78
4 / 8	0	265	90
4 / 9	-55.096598	259.209114	102
4 / 10	-107.78521	242.089546	114
4 / 11	-155.763092	214.389504	126
4 / 12	-196.933379	177.319611	138
4 / 13	-229.496732	132.5	150
4 / 14	-252.029977	81.889504	162
4 / 15	-263.548302	27.700043	174
4 / 16	-263.548302	-27.700043	186
4 / 17	-252.029977	-81.889504	198
4 / 18	-229.496732	-132.5	210

Row / hole	Exact X mm	Exact Y mm	Angle °
4 / 19	-196.933379	-177.319611	222
4 / 20	-155.763092	-214.389504	234
4 / 21	-107.78521	-242.089546	246
4 / 22	-55.096598	-259.209114	258
4 / 23	0	-265	270
4 / 24	55.096598	-259.209114	282
4 / 25	107.78521	-242.089546	294
4 / 26	155.763092	-214.389504	306
4 / 27	196.933379	-177.319611	318
4 / 28	229.496732	-132.5	330
4 / 29	252.029977	-81.889504	342
4 / 30	263.548302	-27.700043	354

Hole diameter 39.559479 mm; centre opening 112 mm; six retained 8 mm no-hole lanes. Coordinate source canonical SHA-256: 954fd39266e55dbaa2c0c10dcc6578b5a94d9247b1ff95194f0a762b58a32042.

Shaft and shaft-support geometry

Design feature	Saved value	Unit	Provenance / governing rule
Lower Shaft Support	450	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. $hB+LB/2$.
Upper Shaft Support	4290	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. $z1+LT/2$.
Support Arm Width	12	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. 0.02D; tangential width, three arms at 30/150/270 degrees.
Support Arm Depth	24	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. 0.04D; axial depth.
Support Housing Diameter	66	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. 1.5ds; central envelope.
Support Housing Height	24	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. 0.04D; central envelope.
Support Arm Radial Start	33	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. 0.75ds; housing surface.
Support Arm Radial End	300	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. $D/2$; shell interface.
Shaft Lower Termination	450	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Lower support centre.
Shaft Upper Termination	4890	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. $HV+D/4$; drive engagement.
Shaft Length	4440	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Upper minus lower termination.
Shaft Area area	1520.530844	mm ²	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. $\pi ds^2/4$.
Shaft Opening Clearance	34	mm	Class C; System-generated recomputed from approved component solids; no upstream feedback.
Support Azimuths angle	30, 150, 270	deg	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Approved three-arm support clocking.
Shaft Diameter	44	mm	Class C; System-generated Class C. Approved shaft diameter.

Support	Centre Z mm	Housing OD mm	Housing height mm
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Support	Centre Z mm	Housing OD mm	Housing height mm
1	450	66	24
2	4290	66	24

Support / arm / azimuth	Bottom / top Z mm	Ordered XY footprint mm
1 / 1 / 30°	438 / 462	31.578838, 11.303848; 262.807621, 144.803848; 256.807621, 155.196152; 25.578838, 21.696152
1 / 2 / 150°	438 / 462	-25.578838, 21.696152; -256.807621, 155.196152; -262.807621, 144.803848; -31.578838, 11.303848
1 / 3 / 270°	438 / 462	-6, -33; -6, -300; 6, -300; 6, -33
2 / 1 / 30°	4278 / 4302	31.578838, 11.303848; 262.807621, 144.803848; 256.807621, 155.196152; 25.578838, 21.696152
2 / 2 / 150°	4278 / 4302	-25.578838, 21.696152; -256.807621, 155.196152; -262.807621, 144.803848; -31.578838, 11.303848
2 / 3 / 270°	4278 / 4302	-6, -33; -6, -300; 6, -300; 6, -33

Skirt, access opening and seal/drive reserved envelopes

Design feature	Saved value	Unit	Provenance / governing rule
Support Height	300	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D/2; skirt below vessel datum.
Skirt Diameter	600	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D; centred cylindrical envelope.
Skirt Access Width	180	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. 0.30D; access opening at azimuth zero.
Skirt Access Height	150	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. 0.25D; access opening.
Skirt Access Elevation	-150	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. -0.25D; centre elevation.
Seal Elevation	4740	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Top pole; coaxial with shaft.
Seal Diameter	88	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. 2ds; reserved housing, not product selection.
Seal Height	60	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. 0.10D above top pole.
Drive Pedestal Diameter	300	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D/2; coaxial reserved envelope.
Drive Pedestal Height	150	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D/4 above top pole.
Drive Body Diameter	300	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D/2; reserved envelope.
Drive Body Height	450	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. 3D/4 above pedestal.
Drive Height	600	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. D including pedestal; not gearbox adequacy.
Skirt Access Azimuth angle	0	deg	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Approved skirt access orientation.

Saved component envelope	Diameter mm	Bottom Z mm	Top Z mm
Saved component envelope	Diameter mm	Bottom Z mm	Top Z mm
Skirt	600	-300	0
Drive-pedestal	300	4740	4890
Drive-body	300	4890	5340
Seal	88	4740	4800
Shaft	44	450	4890

Skirt access width mm	Height mm	Centre Z mm	Azimuth °
180	150	-150	0

Complete connection / nozzle schedule

All connections are preliminary geometric allowances, not standard DN selections or hydraulically qualified piping. Stub OD/projection are saved envelopes. No flange, reinforcement, weld or pipe schedule is implied.

Tag / service	Region / axis	Bore / OD mm	Projection mm	EL / radial offset mm	Azimuth °
P01 RRBO feed	bottom radial	60 / 72	60	510 / 300	0
P02 NMP-rich extract outlet	bottom radial	60 / 72	60	390 / 300	180
P03 NMP feed	top radial	60 / 72	60	4230 / 300	180
P04 RRBO-rich raffinate outlet	top radial	60 / 72	60	4350 / 300	0
A01 Low-shell flush	bottom radial	24 / 28.8	60	270 / 300	270
A02 High-shell equalization	top radial	24 / 28.8	60	4470 / 300	270
S01 Lower sample connection	bottom radial	15 / 18	60	630 / 300	90
S02 Upper sample connection	top radial	15 / 18	60	4110 / 300	90
I01 Lower instrument connection	bottom radial	15 / 18	60	330 / 300	45
I02 Upper instrument connection	top radial	15 / 18	60	4170 / 300	45
D01 Bottom-head drain	bottom-head down	24 / 28.8	60	0 / 0	N/A — axial
V01 Top-head crown-region vent	top-head up	24 / 28.8	60	4710 / 180	315

Tag / axis	Surface centre XYZ mm	External end XYZ mm	Head intersection Z min / max mm
P01 / radial	300, 0, 510	360, 0, 510	N/A — straight shell
P02 / radial	-300, 0, 390	-360, 0, 390	N/A — straight shell
P03 / radial	-300, 0, 4230	-360, 0, 4230	N/A — straight shell
P04 / radial	300, 0, 4350	360, 0, 4350	N/A — straight shell
A01 / radial	0, -300, 270	0, -360, 270	N/A — straight shell
A02 / radial	0, -300, 4470	0, -360, 4470	N/A — straight shell
S01 / radial	0, 300, 630	0, 360, 630	N/A — straight shell
S02 / radial	0, 300, 4110	0, 360, 4110	N/A — straight shell
I01 / radial	212.132034, 212.132034, 330	254.558441, 254.558441, 330	N/A — straight shell
I02 / radial	212.132034, 212.132034, 4170	254.558441, 254.558441, 4170	N/A — straight shell
D01 / down	0, 0, 0	0, 0, -60	0 / 0.1729
V01 / up	127.279221, -127.279221, 4710	127.279221, -127.279221, 4770	4704.24605 / 4715.076617

Tag	Saved connection rationale
Tag	Saved connection rationale
P01	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P02	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P03	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P04	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
A01	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/25; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
A02	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/25; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
S01	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
S02	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
I01	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I02	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
D01	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bottom-pole axial drain; azimuth N/A.
V01	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: $r=0.30D$, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.

Head penetration boundaries: every saved sample point follows in its stored order. Close each boundary loop; these samples describe the saved neck/head intersection, not a new machining allowance.

Connection	Boundary point	Global X, Y, Z mm
D01	1	14.4, 0, 0.1729
D01	2	14.33066, 1.411447, 0.1729
D01	3	14.123308, 2.809301, 0.1729
D01	4	13.779941, 4.180099, 0.1729
D01	5	13.303865, 5.510641, 0.1729
D01	6	12.699666, 6.788113, 0.1729
D01	7	11.973162, 8.000211, 0.1729
D01	8	11.131351, 9.135263, 0.1729
D01	9	10.182338, 10.182338, 0.1729
D01	10	9.135263, 11.131351, 0.1729
D01	11	8.000211, 11.973162, 0.1729
D01	12	6.788113, 12.699666, 0.1729

Connection	Boundary point	Global X, Y, Z mm
D01	13	5.510641, 13.303865, 0.1729
D01	14	4.180099, 13.779941, 0.1729
D01	15	2.809301, 14.123308, 0.1729
D01	16	1.411447, 14.33066, 0.1729
D01	17	0, 14.4, 0.1729
D01	18	-1.411447, 14.33066, 0.1729
D01	19	-2.809301, 14.123308, 0.1729
D01	20	-4.180099, 13.779941, 0.1729
D01	21	-5.510641, 13.303865, 0.1729
D01	22	-6.788113, 12.699666, 0.1729
D01	23	-8.000211, 11.973162, 0.1729
D01	24	-9.135263, 11.131351, 0.1729
D01	25	-10.182338, 10.182338, 0.1729
D01	26	-11.131351, 9.135263, 0.1729
D01	27	-11.973162, 8.000211, 0.1729
D01	28	-12.699666, 6.788113, 0.1729
D01	29	-13.303865, 5.510641, 0.1729
D01	30	-13.779941, 4.180099, 0.1729
D01	31	-14.123308, 2.809301, 0.1729
D01	32	-14.33066, 1.411447, 0.1729
D01	33	-14.4, 0, 0.1729
D01	34	-14.33066, -1.411447, 0.1729
D01	35	-14.123308, -2.809301, 0.1729
D01	36	-13.779941, -4.180099, 0.1729
D01	37	-13.303865, -5.510641, 0.1729
D01	38	-12.699666, -6.788113, 0.1729
D01	39	-11.973162, -8.000211, 0.1729
D01	40	-11.131351, -9.135263, 0.1729
D01	41	-10.182338, -10.182338, 0.1729
D01	42	-9.135263, -11.131351, 0.1729
D01	43	-8.000211, -11.973162, 0.1729
D01	44	-6.788113, -12.699666, 0.1729
D01	45	-5.510641, -13.303865, 0.1729
D01	46	-4.180099, -13.779941, 0.1729
D01	47	-2.809301, -14.123308, 0.1729

Connection	Boundary point	Global X, Y, Z mm
D01	48	-1.411447, -14.33066, 0.1729
D01	49	0, -14.4, 0.1729
D01	50	1.411447, -14.33066, 0.1729
D01	51	2.809301, -14.123308, 0.1729
D01	52	4.180099, -13.779941, 0.1729
D01	53	5.510641, -13.303865, 0.1729
D01	54	6.788113, -12.699666, 0.1729
D01	55	8.000211, -11.973162, 0.1729
D01	56	9.135263, -11.131351, 0.1729
D01	57	10.182338, -10.182338, 0.1729
D01	58	11.131351, -9.135263, 0.1729
D01	59	11.973162, -8.000211, 0.1729
D01	60	12.699666, -6.788113, 0.1729
D01	61	13.303865, -5.510641, 0.1729
D01	62	13.779941, -4.180099, 0.1729
D01	63	14.123308, -2.809301, 0.1729
D01	64	14.33066, -1.411447, 0.1729
V01	1	141.679221, -127.279221, 4705.895425
V01	2	141.609881, -125.867774, 4706.301274
V01	3	141.402529, -124.46992, 4706.739629
V01	4	141.059161, -123.099121, 4707.205929
V01	5	140.583086, -121.768579, 4707.695379
V01	6	139.978887, -120.491108, 4708.203008
V01	7	139.252383, -119.279009, 4708.723725
V01	8	138.410571, -118.143957, 4709.252379
V01	9	137.461558, -117.096883, 4709.783805
V01	10	136.414484, -116.14787, 4710.312885
V01	11	135.279432, -115.306058, 4710.834586
V01	12	134.067334, -114.579554, 4711.344011
V01	13	132.789862, -113.975355, 4711.836438
V01	14	131.45932, -113.49928, 4712.307359
V01	15	130.088521, -113.155913, 4712.752511
V01	16	128.690667, -112.948561, 4713.167909
V01	17	127.279221, -112.879221, 4713.54987
V01	18	125.867774, -112.948561, 4713.895042

Connection	Boundary point	Global X, Y, Z mm
V01	19	124.46992, -113.155913, 4714.200421
V01	20	123.099121, -113.49928, 4714.463368
V01	21	121.768579, -113.975355, 4714.681626
V01	22	120.491108, -114.579554, 4714.853333
V01	23	119.279009, -115.306058, 4714.977029
V01	24	118.143957, -116.14787, 4715.051667
V01	25	117.096883, -117.096883, 4715.076617
V01	26	116.14787, -118.143957, 4715.051667
V01	27	115.306058, -119.279009, 4714.977029
V01	28	114.579554, -120.491108, 4714.853333
V01	29	113.975355, -121.768579, 4714.681626
V01	30	113.49928, -123.099121, 4714.463368
V01	31	113.155913, -124.46992, 4714.200421
V01	32	112.948561, -125.867774, 4713.895042
V01	33	112.879221, -127.279221, 4713.54987
V01	34	112.948561, -128.690667, 4713.167909
V01	35	113.155913, -130.088521, 4712.752511
V01	36	113.49928, -131.45932, 4712.307359
V01	37	113.975355, -132.789862, 4711.836438
V01	38	114.579554, -134.067334, 4711.344011
V01	39	115.306058, -135.279432, 4710.834586
V01	40	116.14787, -136.414484, 4710.312885
V01	41	117.096883, -137.461558, 4709.783805
V01	42	118.143957, -138.410571, 4709.252379
V01	43	119.279009, -139.252383, 4708.723725
V01	44	120.491108, -139.978887, 4708.203008
V01	45	121.768579, -140.583086, 4707.695379
V01	46	123.099121, -141.059161, 4707.205929
V01	47	124.46992, -141.402529, 4706.739629
V01	48	125.867774, -141.609881, 4706.301274
V01	49	127.279221, -141.679221, 4705.895425
V01	50	128.690667, -141.609881, 4705.526354
V01	51	130.088521, -141.402529, 4705.197983
V01	52	131.45932, -141.059161, 4704.913837
V01	53	132.789862, -140.583086, 4704.67699

Connection	Boundary point	Global X, Y, Z mm
V01	54	134.067334, -139.978887, 4704.490023
V01	55	135.279432, -139.252383, 4704.354984
V01	56	136.414484, -138.410571, 4704.273359
V01	57	137.461558, -137.461558, 4704.24605
V01	58	138.410571, -136.414484, 4704.273359
V01	59	139.252383, -135.279432, 4704.354984
V01	60	139.978887, -134.067334, 4704.490023
V01	61	140.583086, -132.789862, 4704.67699
V01	62	141.059161, -131.45932, 4704.913837
V01	63	141.402529, -130.088521, 4705.197983
V01	64	141.609881, -128.690667, 4705.526354

Component quantities and complete elevation schedules

Every saved rotor, stator and compartment is listed. Stator plate thickness is centred on each boundary plane. These are not representative or truncated schedules.

Component	Quantity	Diameter mm	Thickness / axial size mm
Approved double-entry radial-flow shrouded six-blade turbine	18	198	32
Approved perforated stator: central opening plus 84 exact profile holes	19	600	4
Continuous shaft envelope; mechanical qualification excluded	1	44	N/A — solid shaft envelope
Three-arm shaft supports, azimuths 30/150/270 deg	2	600	24

Other modelled components	Quantity / definition
Head envelopes	2
Shaft-support assemblies	2
Support arms	6
Connections	12
Blades per rotor	6
Reserved component envelopes	Skirt, Drive-pedestal, Drive-body, Seal, Shaft
Skirt access opening	1

Compartment	Bottom Z mm	Top Z mm	Rotor centre Z mm	Saved stator Z mm
1	750	930	840	930
2	930	1110	1020	1110
3	1110	1290	1200	1290
4	1290	1470	1380	1470
5	1470	1650	1560	1650
6	1650	1830	1740	1830
7	1830	2010	1920	2010
8	2010	2190	2100	2190
9	2190	2370	2280	2370
10	2370	2550	2460	2550
11	2550	2730	2640	2730
12	2730	2910	2820	2910
13	2910	3090	3000	3090
14	3090	3270	3180	3270
15	3270	3450	3360	3450
16	3450	3630	3540	3630

Compartment	Bottom Z mm	Top Z mm	Rotor centre Z mm	Saved stator Z mm
17	3630	3810	3720	3810
18	3810	3990	3900	3990

Approved double-entry radial-flow shrouded six-blade turbine — every saved elevation

Component ID	Ordinal	Centre / boundary Z mm
R	1	840
R	2	1020
R	3	1200
R	4	1380
R	5	1560
R	6	1740
R	7	1920
R	8	2100
R	9	2280
R	10	2460
R	11	2640
R	12	2820
R	13	3000
R	14	3180
R	15	3360
R	16	3540
R	17	3720
R	18	3900

Approved perforated stator: central opening plus 84 exact profile holes — every saved elevation

Component ID	Ordinal	Centre / boundary Z mm
S	1	750
S	2	930
S	3	1110
S	4	1290
S	5	1470
S	6	1650
S	7	1830
S	8	2010
S	9	2190

Component ID	Ordinal	Centre / boundary Z mm
S	10	2370
S	11	2550
S	12	2730
S	13	2910
S	14	3090
S	15	3270
S	16	3450
S	17	3630
S	18	3810
S	19	3990

Continuous shaft envelope; mechanical qualification excluded — every saved elevation

Component ID	Ordinal	Centre / boundary Z mm
SH	1	450
SH	2	4890

Three-arm shaft supports, azimuths 30/150/270 deg — every saved elevation

Component ID	Ordinal	Centre / boundary Z mm
SUP	1	450
SUP	2	4290

Critical clearances

The saved clearances are geometric checks only; manufacturing tolerances, thermal movement and mechanical allowances are not introduced by this report.

Design feature	Saved value	Unit	Provenance / governing rule
Rotor Wall Clearance	201	mm	Class C; System-generated recomputed from approved component solids; no upstream feedback.
Shaft Opening Clearance	34	mm	Class C; System-generated recomputed from approved component solids; no upstream feedback.
Stator Radial Width	244	mm	Class C; System-generated recomputed from approved component solids; no upstream feedback.
Rotor Lower Clearance	72	mm	Class C; System-generated recomputed from approved component solids; no upstream feedback.
Rotor Upper Clearance	72	mm	Class C; System-generated recomputed from approved component solids; no upstream feedback.

Validation results — complete saved check register

298 passed of 298. Status is from this saved revision, not a rerun against current upstream results.

Check	Result	Validation requirement
Authority	PASS	Frozen sources must be identified, current and compatible.
Column Diameter	PASS	columnDiameterM must be finite and positive.
Rotor Diameter	PASS	rotorDiameterM must be finite and positive.
Rotor Diameter Ratio	PASS	rotorDiameterRatio must be finite and positive.
Compartment Height	PASS	compartmentHeightM must be finite and positive.
Compartment Count	PASS	compartmentCount must be finite and positive.
Installed Active Height	PASS	installedActiveHeightM must be finite and positive.
Required Active Height	PASS	requiredActiveHeightM must be finite and positive.
Design Nt	PASS	designNt must be finite and positive.
Selected Rpm	PASS	selectedRpm must be finite and positive.
Rpm Min	PASS	rpmMin must be finite and positive.
Rpm Max	PASS	rpmMax must be finite and positive.
Stator Free Area Ratio	PASS	statorFreeAreaRatio must be finite and positive.
Count	PASS	Positive integer count; drawing capacity 1000 compartments.
Stack	PASS	Frozen count × pitch must equal frozen installed height; no repair.
Sizing-method	PASS	Frozen adopted sizing basis must be Nt=7 and compartment efficiency=0.40.
Count-efficiency	PASS	Frozen count must equal ceil(Nt / adopted compartment efficiency).
Height	PASS	Frozen required and installed heights must both equal count × pitch.
Implied-hets-diagnostic	PASS	Implied installed HETS must equal installed height / Nt and remains diagnostic only.
Rotor-ratio	PASS	Frozen rotor diameter/ratio must agree and fit column.
Rpm	PASS	Frozen selected RPM must lie inside frozen window.
Phase	PASS	Frozen phase continuity is required.
Phi-domain	PASS	Empirical free-area input must be between zero and one.
Finite-column Diameter	PASS	columnDiameterM must be finite.
Finite-compartment Height	PASS	compartmentHeightM must be finite.
Finite-rotor Diameter Ratio	PASS	rotorDiameterRatio must be finite.
Finite-rotor Diameter	PASS	rotorDiameterM must be finite.
Finite-stator Free Area Ratio	PASS	statorFreeAreaRatio must be finite.
Finite-selected Rpm	PASS	selectedRpm must be finite.
Finite-rpm Min	PASS	rpmMin must be finite.
Finite-rpm Max	PASS	rpmMax must be finite.
Finite-compartment Count	PASS	compartmentCount must be finite.

Check	Result	Validation requirement
Finite-required Active Height	PASS	requiredActiveHeightM must be finite.
Finite-installed Active Height	PASS	installedActiveHeightM must be finite.
Finite-design Nt	PASS	designNt must be finite.
Finite-design Compartment Efficiency	PASS	designCompartmentEfficiency must be finite.
Finite-implied Installed Hets MPer Theoretical Stage	PASS	impliedInstalledHetsMPerTheoreticalStage must be finite.
Finite-shaft Diameter	PASS	shaftDiameterM must be finite.
Finite-stator Thickness	PASS	statorThicknessM must be finite.
Finite-blade Count	PASS	bladeCount must be finite.
Finite-blade Height	PASS	bladeHeightM must be finite.
Finite-hub Diameter	PASS	hubDiameterM must be finite.
Finite-hub Height	PASS	hubHeightM must be finite.
Finite-blade Thickness	PASS	bladeThicknessM must be finite.
Finite-blade Radial Length	PASS	bladeRadialLengthM must be finite.
Finite-blade Inner Segment Length	PASS	bladeInnerSegmentLengthM must be finite.
Finite-blade Outer Segment Length	PASS	bladeOuterSegmentLengthM must be finite.
Finite-blade Inner Segment Height	PASS	bladeInnerSegmentHeightM must be finite.
Finite-blade Outer Segment Height	PASS	bladeOuterSegmentHeightM must be finite.
Finite-rotor Offset	PASS	rotorOffsetM must be finite.
Finite-bottom Disengagement	PASS	bottomDisengagementM must be finite.
Finite-top Disengagement	PASS	topDisengagementM must be finite.
Finite-bottom Head Depth	PASS	bottomHeadDepthM must be finite.
Finite-top Head Depth	PASS	topHeadDepthM must be finite.
Finite-head Straight Flange	PASS	headStraightFlangeM must be finite.
Finite-support Height	PASS	supportHeightM must be finite.
Finite-skirt Diameter	PASS	skirtDiameterM must be finite.
Finite-skirt Access Width	PASS	skirtAccessWidthM must be finite.
Finite-skirt Access Height	PASS	skirtAccessHeightM must be finite.
Finite-skirt Access Elevation	PASS	skirtAccessElevationM must be finite.
Finite-lower Shaft Support	PASS	lowerShaftSupportM must be finite.
Finite-upper Shaft Support	PASS	upperShaftSupportM must be finite.
Finite-support Arm Width	PASS	supportArmWidthM must be finite.
Finite-support Arm Depth	PASS	supportArmDepthM must be finite.

Check	Result	Validation requirement
Finite-support Housing Diameter	PASS	supportHousingDiameterM must be finite.
Finite-support Housing Height	PASS	supportHousingHeightM must be finite.
Finite-support Arm Radial Start	PASS	supportArmRadialStartM must be finite.
Finite-support Arm Radial End	PASS	supportArmRadialEndM must be finite.
Finite-shaft Lower Termination	PASS	shaftLowerTerminationM must be finite.
Finite-shaft Upper Termination	PASS	shaftUpperTerminationM must be finite.
Finite-shaft Length	PASS	shaftLengthM must be finite.
Finite-seal Elevation	PASS	sealElevationM must be finite.
Finite-seal Diameter	PASS	sealDiameterM must be finite.
Finite-seal Height	PASS	sealHeightM must be finite.
Finite-drive Pedestal Diameter	PASS	drivePedestalDiameterM must be finite.
Finite-drive Pedestal Height	PASS	drivePedestalHeightM must be finite.
Finite-drive Body Diameter	PASS	driveBodyDiameterM must be finite.
Finite-drive Body Height	PASS	driveBodyHeightM must be finite.
Finite-drive Height	PASS	driveHeightM must be finite.
Finite-stator Opening Diameter	PASS	statorOpeningDiameterM must be finite.
Finite-column Area area	PASS	columnAreaM2 must be finite.
Finite-shaft Area area	PASS	shaftAreaM2 must be finite.
Finite-stator Gross Open Area area	PASS	statorGrossOpenAreaM2 must be finite.
Finite-stator Net Open Area area	PASS	statorNetOpenAreaM2 must be finite.
Finite-gross Free Area Ratio	PASS	grossFreeAreaRatio must be finite.
Finite-shaft Blocked Free Area Ratio	PASS	shaftBlockedFreeAreaRatio must be finite.
Finite-calculated Free Area Ratio	PASS	calculatedFreeAreaRatio must be finite.
Finite-rotor Wall Clearance	PASS	rotorWallClearanceM must be finite.
Finite-shaft Opening Clearance	PASS	shaftOpeningClearanceM must be finite.
Finite-stator Radial Width	PASS	statorRadialWidthM must be finite.
Finite-rotor Axial Envelope	PASS	rotorAxialEnvelopeM must be finite.
Finite-rotor Lower Clearance	PASS	rotorLowerClearanceM must be finite.
Finite-rotor Upper Clearance	PASS	rotorUpperClearanceM must be finite.
Finite-active Start	PASS	activeStartM must be finite.
Finite-active End	PASS	activeEndM must be finite.
Finite-top Tangent	PASS	topTangentM must be finite.
Finite-vessel Height	PASS	vesselHeightM must be finite.
Finite-overall Height	PASS	overallHeightM must be finite.

Check	Result	Validation requirement
Finite-skirt Access Azimuth angle	PASS	skirtAccessAzimuthDeg must be finite.
Finite-P01-elevation	PASS	P01-elevationM must be finite.
Finite-P01-bore	PASS	P01-boreM must be finite.
Finite-P01-azimuth angle	PASS	P01-azimuthDeg must be finite.
Finite-P01-radial Offset	PASS	P01-radialOffsetM must be finite.
Finite-P01-outside Diameter	PASS	P01-outsideDiameterM must be finite.
Finite-P01-projection	PASS	P01-projectionM must be finite.
Finite-P02-elevation	PASS	P02-elevationM must be finite.
Finite-P02-bore	PASS	P02-boreM must be finite.
Finite-P02-azimuth angle	PASS	P02-azimuthDeg must be finite.
Finite-P02-radial Offset	PASS	P02-radialOffsetM must be finite.
Finite-P02-outside Diameter	PASS	P02-outsideDiameterM must be finite.
Finite-P02-projection	PASS	P02-projectionM must be finite.
Finite-P03-elevation	PASS	P03-elevationM must be finite.
Finite-P03-bore	PASS	P03-boreM must be finite.
Finite-P03-azimuth angle	PASS	P03-azimuthDeg must be finite.
Finite-P03-radial Offset	PASS	P03-radialOffsetM must be finite.
Finite-P03-outside Diameter	PASS	P03-outsideDiameterM must be finite.
Finite-P03-projection	PASS	P03-projectionM must be finite.
Finite-P04-elevation	PASS	P04-elevationM must be finite.
Finite-P04-bore	PASS	P04-boreM must be finite.
Finite-P04-azimuth angle	PASS	P04-azimuthDeg must be finite.
Finite-P04-radial Offset	PASS	P04-radialOffsetM must be finite.
Finite-P04-outside Diameter	PASS	P04-outsideDiameterM must be finite.
Finite-P04-projection	PASS	P04-projectionM must be finite.
Finite-A01-elevation	PASS	A01-elevationM must be finite.
Finite-A01-bore	PASS	A01-boreM must be finite.
Finite-A01-azimuth angle	PASS	A01-azimuthDeg must be finite.
Finite-A01-radial Offset	PASS	A01-radialOffsetM must be finite.
Finite-A01-outside Diameter	PASS	A01-outsideDiameterM must be finite.
Finite-A01-projection	PASS	A01-projectionM must be finite.
Finite-A02-elevation	PASS	A02-elevationM must be finite.
Finite-A02-bore	PASS	A02-boreM must be finite.
Finite-A02-azimuth angle	PASS	A02-azimuthDeg must be finite.
Finite-A02-radial Offset	PASS	A02-radialOffsetM must be finite.

Check	Result	Validation requirement
Finite-A02-outside Diameter	PASS	A02-outsideDiameterM must be finite.
Finite-A02-projection	PASS	A02-projectionM must be finite.
Finite-S01-elevation	PASS	S01-elevationM must be finite.
Finite-S01-bore	PASS	S01-boreM must be finite.
Finite-S01-azimuth angle	PASS	S01-azimuthDeg must be finite.
Finite-S01-radial Offset	PASS	S01-radialOffsetM must be finite.
Finite-S01-outside Diameter	PASS	S01-outsideDiameterM must be finite.
Finite-S01-projection	PASS	S01-projectionM must be finite.
Finite-S02-elevation	PASS	S02-elevationM must be finite.
Finite-S02-bore	PASS	S02-boreM must be finite.
Finite-S02-azimuth angle	PASS	S02-azimuthDeg must be finite.
Finite-S02-radial Offset	PASS	S02-radialOffsetM must be finite.
Finite-S02-outside Diameter	PASS	S02-outsideDiameterM must be finite.
Finite-S02-projection	PASS	S02-projectionM must be finite.
Finite-I01-elevation	PASS	I01-elevationM must be finite.
Finite-I01-bore	PASS	I01-boreM must be finite.
Finite-I01-azimuth angle	PASS	I01-azimuthDeg must be finite.
Finite-I01-radial Offset	PASS	I01-radialOffsetM must be finite.
Finite-I01-outside Diameter	PASS	I01-outsideDiameterM must be finite.
Finite-I01-projection	PASS	I01-projectionM must be finite.
Finite-I02-elevation	PASS	I02-elevationM must be finite.
Finite-I02-bore	PASS	I02-boreM must be finite.
Finite-I02-azimuth angle	PASS	I02-azimuthDeg must be finite.
Finite-I02-radial Offset	PASS	I02-radialOffsetM must be finite.
Finite-I02-outside Diameter	PASS	I02-outsideDiameterM must be finite.
Finite-I02-projection	PASS	I02-projectionM must be finite.
Finite-D01-elevation	PASS	D01-elevationM must be finite.
Finite-D01-bore	PASS	D01-boreM must be finite.
Finite-D01-radial Offset	PASS	D01-radialOffsetM must be finite.
Finite-D01-outside Diameter	PASS	D01-outsideDiameterM must be finite.
Finite-D01-projection	PASS	D01-projectionM must be finite.
Finite-D01-surface Edge Elevation Min	PASS	D01-surfaceEdgeElevationMinM must be finite.
Finite-D01-surface Edge Elevation Max	PASS	D01-surfaceEdgeElevationMaxM must be finite.
Head-V01-radial-fit	PASS	Entire neck cylinder fits ellipsoidal head.

Check	Result	Validation requirement
Finite-V01-elevation	PASS	V01-elevationM must be finite.
Finite-V01-bore	PASS	V01-boreM must be finite.
Finite-V01-azimuth angle	PASS	V01-azimuthDeg must be finite.
Finite-V01-radial Offset	PASS	V01-radialOffsetM must be finite.
Finite-V01-outside Diameter	PASS	V01-outsideDiameterM must be finite.
Finite-V01-projection	PASS	V01-projectionM must be finite.
Finite-V01-surface Edge Elevation Min	PASS	V01-surfaceEdgeElevationMinM must be finite.
Finite-V01-surface Edge Elevation Max	PASS	V01-surfaceEdgeElevationMaxM must be finite.
Finite-compartment1-bottom	PASS	compartment1-bottomM must be finite.
Finite-compartment1-top	PASS	compartment1-topM must be finite.
Finite-compartment1-rotor	PASS	compartment1-rotorM must be finite.
Finite-compartment2-bottom	PASS	compartment2-bottomM must be finite.
Finite-compartment2-top	PASS	compartment2-topM must be finite.
Finite-compartment2-rotor	PASS	compartment2-rotorM must be finite.
Finite-compartment3-bottom	PASS	compartment3-bottomM must be finite.
Finite-compartment3-top	PASS	compartment3-topM must be finite.
Finite-compartment3-rotor	PASS	compartment3-rotorM must be finite.
Finite-compartment4-bottom	PASS	compartment4-bottomM must be finite.
Finite-compartment4-top	PASS	compartment4-topM must be finite.
Finite-compartment4-rotor	PASS	compartment4-rotorM must be finite.
Finite-compartment5-bottom	PASS	compartment5-bottomM must be finite.
Finite-compartment5-top	PASS	compartment5-topM must be finite.
Finite-compartment5-rotor	PASS	compartment5-rotorM must be finite.
Finite-compartment6-bottom	PASS	compartment6-bottomM must be finite.
Finite-compartment6-top	PASS	compartment6-topM must be finite.
Finite-compartment6-rotor	PASS	compartment6-rotorM must be finite.
Finite-compartment7-bottom	PASS	compartment7-bottomM must be finite.
Finite-compartment7-top	PASS	compartment7-topM must be finite.
Finite-compartment7-rotor	PASS	compartment7-rotorM must be finite.
Finite-compartment8-bottom	PASS	compartment8-bottomM must be finite.
Finite-compartment8-top	PASS	compartment8-topM must be finite.
Finite-compartment8-rotor	PASS	compartment8-rotorM must be finite.
Finite-compartment9-bottom	PASS	compartment9-bottomM must be finite.
Finite-compartment9-top	PASS	compartment9-topM must be finite.

Check	Result	Validation requirement
Finite-compartment9-rotor	PASS	compartment9-rotorM must be finite.
Finite-compartment10-bottom	PASS	compartment10-bottomM must be finite.
Finite-compartment10-top	PASS	compartment10-topM must be finite.
Finite-compartment10-rotor	PASS	compartment10-rotorM must be finite.
Finite-compartment11-bottom	PASS	compartment11-bottomM must be finite.
Finite-compartment11-top	PASS	compartment11-topM must be finite.
Finite-compartment11-rotor	PASS	compartment11-rotorM must be finite.
Finite-compartment12-bottom	PASS	compartment12-bottomM must be finite.
Finite-compartment12-top	PASS	compartment12-topM must be finite.
Finite-compartment12-rotor	PASS	compartment12-rotorM must be finite.
Finite-compartment13-bottom	PASS	compartment13-bottomM must be finite.
Finite-compartment13-top	PASS	compartment13-topM must be finite.
Finite-compartment13-rotor	PASS	compartment13-rotorM must be finite.
Finite-compartment14-bottom	PASS	compartment14-bottomM must be finite.
Finite-compartment14-top	PASS	compartment14-topM must be finite.
Finite-compartment14-rotor	PASS	compartment14-rotorM must be finite.
Finite-compartment15-bottom	PASS	compartment15-bottomM must be finite.
Finite-compartment15-top	PASS	compartment15-topM must be finite.
Finite-compartment15-rotor	PASS	compartment15-rotorM must be finite.
Finite-compartment16-bottom	PASS	compartment16-bottomM must be finite.
Finite-compartment16-top	PASS	compartment16-topM must be finite.
Finite-compartment16-rotor	PASS	compartment16-rotorM must be finite.
Finite-compartment17-bottom	PASS	compartment17-bottomM must be finite.
Finite-compartment17-top	PASS	compartment17-topM must be finite.
Finite-compartment17-rotor	PASS	compartment17-rotorM must be finite.
Finite-compartment18-bottom	PASS	compartment18-bottomM must be finite.
Finite-compartment18-top	PASS	compartment18-topM must be finite.
Finite-compartment18-rotor	PASS	compartment18-rotorM must be finite.
Drive-seal-fit	PASS	Seal reserved envelope fits drive pedestal; shaft deliberately engages seal, supports and drive.
Skirt-drain	PASS	Drain neck fits inside skirt and remains above ground.
P01-region-envelope	PASS	Full neck OD lies in assigned straight-shell zone.
P02-region-envelope	PASS	Full neck OD lies in assigned straight-shell zone.
P03-region-envelope	PASS	Full neck OD lies in assigned straight-shell zone.
P04-region-envelope	PASS	Full neck OD lies in assigned straight-shell zone.
A01-region-envelope	PASS	Full neck OD lies in assigned straight-shell zone.

Check	Result	Validation requirement
A02-region-envelope	PASS	Full neck OD lies in assigned straight-shell zone.
S01-region-envelope	PASS	Full neck OD lies in assigned straight-shell zone.
S02-region-envelope	PASS	Full neck OD lies in assigned straight-shell zone.
I01-region-envelope	PASS	Full neck OD lies in assigned straight-shell zone.
I02-region-envelope	PASS	Full neck OD lies in assigned straight-shell zone.
P01-P02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P01-P03-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P01-P04-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P01-A01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P01-A02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P01-S01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P01-S02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P01-I01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P01-I02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P02-P03-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P02-P04-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P02-A01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P02-A02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P02-S01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P02-S02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P02-I01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P02-I02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P03-P04-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P03-A01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P03-A02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P03-S01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P03-S02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P03-I01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P03-I02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P04-A01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P04-A02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P04-S01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P04-S02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P04-I01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
P04-I02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).

Check	Result	Validation requirement
A01-A02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
A01-S01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
A01-S02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
A01-I01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
A01-I02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
A02-S01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
A02-S02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
A02-I01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
A02-I02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
S01-S02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
S01-I01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
S01-I02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
S02-I01-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
S02-I02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
I01-I02-stub-envelope	PASS	Complete outward neck envelopes are disjoint (not merely bore openings).
Vent-drive-seal-envelope	PASS	Vertical vent cylinder clears full drive/pedestal/seal radial envelopes.
Head-connections-internals	PASS	Actual ellipsoid/neck intersection envelopes clear internal supports and shaft lower termination.
Head-shell-connections	PASS	Head connections and straight-shell stubs are axially disjoint.
Approved-hole-count	PASS	Exactly 84 manifest hole centres retained.
Approved-gross-area	PASS	Computed centre plus 84-hole physical area divided by computed column area reproduces gross free-area fraction 0.40.
Approved-shaft-centre-clearance	PASS	Approved shaft clears the approved central opening.
Approved-hub-fit	PASS	Approved shaft, hub and rotor diameters are nested.
Approved-blade-topology	PASS	Six exact cylindrical-trim blade solids do not overlap at the hub.
Approved-axial-clearance	PASS	Approved 32 mm rotor envelope clears the 4 mm boundary plates within the inherited 180 mm pitch.
Approved-nozzle-stator-envelope	PASS	Saved radial nozzle OD envelopes are axially disjoint from all approved 4 mm stator plates; external interfaces remain on hold.
Approved-support-bracketing	PASS	Saved support envelopes geometrically bracket the approved stack; support-wall and shaft interfaces remain preliminary holds.
Approved-hole-topology	PASS	All exact holes are mutually disjoint and clear the centre opening and plate perimeter.

Approved-component rules, source hashes and retained lineage

These identifiers belong to the saved revision. Missing legacy metadata is explicitly reported, never substituted with current sources.

Stage 3 saved reference	Value
Id	64
Result / Engine / Id	ecr_stage3_stage4_optimizer
Result / Engine / Version	ECR_STAGE3_STAGE4_OPTIMIZER_V1.3.0
Result / Engine / Implementation Hash	faa56fdd99b0904a5e524fb8350859192d94b269331e91bbd34cc5d946ccb550
Result / Process Basis / Schema Version	ECR_PRE_PILOT_HYDRODYNAMIC_PROCESS_BASIS_V1
Result / Process Basis / Stage1 Snapshot Hash	549dffa428356ee5e94d8baff297c114e1f0c6f345b84e121181e7b5f6b5f15b
Result / Schema Version	ECR_STAGE3_STAGE4_OPTIMIZER_RESULT_V1
Result / Calculation Hash	f7cacea88792a9b70aecb65e187728c5e4aae1fa038920a450d9e2385bf850ef
Result / Stage1 Authority / Snapshot Hash	549dffa428356ee5e94d8baff297c114e1f0c6f345b84e121181e7b5f6b5f15b
Result / Stage1 Authority / Process Basis Schema Version	ECR_PRE_PILOT_HYDRODYNAMIC_PROCESS_BASIS_V1
Result / Stage4 Geometry Input / Optimizer Result Hash	8616a167dde2a52f5cd14c65e132776e9796d2f360c204dbcff4903991fc9c4e
Immutable Hash	e2afc059444f3c22c5d3a64b83ed81b59b5a122bd51ac1c9bc9897337d788249

Stage 4 saved reference	Value
Id	15
Result / Implementation / Version	ECR_STAGE4_ADOPTED_COMPARTMENT EFFICIENCY_V1_FIXED_DESIGN_NT7_ETA0.40
Result / Implementation / Implementation Hash	e3f597aec0f622019ff9526e80ca4c4104b761179ac41ce9b4019112ac2473c8
Result / Current Optimizer / Version	ECR_STAGE3_STAGE4_OPTIMIZER_V1.3.0
Result / Current Optimizer / Result Hash	8616a167dde2a52f5cd14c65e132776e9796d2f360c204dbcff4903991fc9c4e
Result / Current Optimizer / Implementation Hash	faa56fdd99b0904a5e524fb8350859192d94b269331e91bbd34cc5d946ccb550
Result / Actual Stage2 Nt Reference / Stage2 Job Id	145e962e-892a-4424-aab9-91532af9f5ef
Result / Actual Stage2 Nt Reference / Stage2 Result Hash	4a5c37ac721bc37616b56d706a182d492bb60d56c281a48257ecc195db5d06ef
Result / Selected Stage3 Hydraulics / Stage3 Run Id	64
Result / Selected Stage3 Hydraulics / Optimizer Version	ECR_STAGE3_STAGE4_OPTIMIZER_V1.3.0
Result / Selected Stage3 Hydraulics / Optimizer Result Hash	8616a167dde2a52f5cd14c65e132776e9796d2f360c204dbcff4903991fc9c4e

Stage 4 saved reference	Value
Result / Selected Stage3 Hydraulics / Stage3 Immutable Hash	e2afc059444f3c22c5d3a64b83ed81b59b5a122bd51ac1c9bc9897337d788249
Result / Selected Stage3 Hydraulics / Optimizer Implementation Hash	faa56fdd99b0904a5e524fb8350859192d94b269331e91bbd34cc5d946ccb550
Result / Stage2 Stage1 Compatibility / Current Stage1 Snapshot Hash	549dfa428356ee5e94d8baff297c114e1f0c6f345b84e121181e7b5f6b5f15b
Result / Stage2 Stage1 Compatibility / Persisted Stage2 Stage1 Snapshot Hash	6ccf5381167d4deb01d313881b63f5bb537c8bc729e3902b2919fc1f4397fbe9
Result / Stage2 Stage1 Compatibility / Current Equilibrium Scientific Input Hash	faa75060eb246fd90993e01245e4e052b968c7bf81bb262532abfea2ac7ce1af
Result / Stage2 Stage1 Compatibility / Persisted Equilibrium Scientific Input Hash	faa75060eb246fd90993e01245e4e052b968c7bf81bb262532abfea2ac7ce1af
Lineage Hash	c694966f243f50a696df8fd0143bfd365f9148939e0e1db6967c27bfe051dcde

Frozen rules manifest	Value
Id	ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R3_APPROVED_TURBINE_PERFORATED_STATOR
Rules / Step	inner Lb/2 height HR/2; outer Lb/2 height HR; symmetric axial profile; azimuths 0,60,120,180,240,300
Rules / Drive	sealOD2ds height.10D at HV; pedestalOD D/2 heightD/4; bodyOD D/2 height3D/4; totalheightD
Rules / Heads	2:1 ellipse; radial semiaxis D/2; axial semiaxis D/4; straight flange zero
Rules / Rotor	D198 double-entry shrouded turbine; eye130; shaft44; hub70x24; six t3 blades; web16; paddle28; upper/lower t2 shrouds; overall32
Rules / Shaft	ds=(10/45)DR; lower=hB+LB/2; upper=HV+D/4
Rules / Stack	N=ceil(7/0.40)=18; HA=3.24 m; N+1=19 stators; inherited pitch 0.18 m
Rules / Stator	D600 t4 plate; centre opening112 from CE9; 84 exact D39.559479027818114 holes at approved manifest coordinates
Rules / Opening	do=D*sqrt(phi_s); gross=(do/D)^2; net=(do^2-ds^2)/D^2; no feedback
Rules / Routing	NMP down/RRBO up; frozen phase continuity unchanged
Rules / Endzones	LB=LT=D; no residence-time claim
Rules / Schedule	P01 lower .60 0; P02 lower .40 180; P03 upper .40 180; P04 upper .60 0; A01 lower .20 270; A02 upper .80 270; S01 lower .80 90; S02 upper .20 90; I01 lower .30 45; I02 upper .30 45; D01 bottom pole down; V01 top head r=.30D azimuth315 up
Rules / Supports	skirt D diameter D/2 height; access .30D by .25D at z=-.25D, azimuth0; supports at hB+LB/2,z1+LT/2; arms azimuth30,150,270 width.02D depth.04D span .75ds to D/2; housingOD1.5ds height.04D
Rules / Connections	process bore D/10; drain/vent/flush/equalization D/25; sample/instrument D/40; OD1.20bore; projectionD/10
Rules / Intersections	Exact ellipsoid/cylinder boundary; full OD neck and projected envelope clearances; intended shaft/hub/support/drive interfaces excluded from clashes
Units	m, m², rpm, degrees; vessel bottom datum

Frozen rules manifest	Value
Evidence / A	Frozen Stage3/4 authority
Evidence / B	Documented reference construction family
Evidence / C	Approved R1 engineering geometry
Revision	3
Authority	User-approved component manifest plus frozen Stage-3/4 authority; exact D600 template, no scaling
Exclusions	Fabrication, structural and shaft-dynamic qualification, hydraulic nozzle sizing and disengagement performance; no user-entered dimensions
Approved Component Manifest / Record Type	USER_APPROVED_PRE_PILOT_COMPONENT_BASIS
Approved Component Manifest / Source Design Id	269
Approved Component Manifest / Canonical Sha256	954fd39266e55dbaa2c0c10dcc6578b5a94d9247b1ff95194f0a762b58a32042
Approved Component Manifest / Source Stage3 Run Id	64
Approved Component Manifest / Source Stage4 Calculation Id	13

Additional saved design selections and evidence

Supplementary saved parameters not repeated in the component sections. All dimensions retain their saved meaning.

Design feature	Saved value	Unit	Provenance / governing rule
Source Hash	04dfb56f2139741bb5813460c9a575e95b6927718f837ed09629fb70e2bbb610	—	Class A; Inherited ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Sizing Method	ADOPTED_COMPARTMENT_EFFICIENCY	—	Class A; Inherited ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Design Compartment Efficiency	0.4	—	Class A; Inherited ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Implied Installed Hets MPer Theoretical Stage	0.462857	—	Class A; Inherited ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class A. Exact frozen Stage-3/4 authority; scientific status unchanged.
Flow Arrangement	nmp-down-rrbo-up	—	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. Explicit application layout; continuity and upstream hydraulics unchanged.
Design Envelope	hc/D 0.20–0.30; DR/D 0.33–0.50; phi_s 0.20–0.40	—	Class B; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class B. Perry 8e p15-83 → Pratt-Stevens; advisory, never clamp inherited geometry.
P01-elevation	510	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P01-bore	60	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P01-azimuth angle	0	deg	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P01-radial Offset	300	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P01-outside Diameter	72	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.

Design feature	Saved value	Unit	Provenance / governing rule
P01-projection	60	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P02-elevation	390	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P02-bore	60	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P02-azimuth angle	180	deg	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P02-radial Offset	300	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P02-outside Diameter	72	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P02-projection	60	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P03-elevation	4230	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P03-bore	60	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P03-azimuth angle	180	deg	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.

Design feature	Saved value	Unit	Provenance / governing rule
P03-radial Offset	300	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P03-outside Diameter	72	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P03-projection	60	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.4; geometric allowance, not hydraulic/DN sizing.
P04-elevation	4350	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P04-bore	60	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P04-azimuth angle	0	deg	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P04-radial Offset	300	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P04-outside Diameter	72	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
P04-projection	60	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/10; zone fraction 0.6; geometric allowance, not hydraulic/DN sizing.
A01-elevation	270	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/25; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.

Design feature	Saved value	Unit	Provenance / governing rule
A01-bore	24	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/25; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
A01-azimuth angle	270	deg	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/25; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
A01-radial Offset	300	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/25; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
A01-outside Diameter	28.8	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/25; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
A01-projection	60	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/25; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
A02-elevation	4470	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/25; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
A02-bore	24	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/25; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
A02-azimuth angle	270	deg	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/25; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
A02-radial Offset	300	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/25; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
A02-outside Diameter	28.8	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/25; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.

Design feature	Saved value	Unit	Provenance / governing rule
A02-projection	60	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/25; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
S01-elevation	630	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
S01-bore	15	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
S01-azimuth angle	90	deg	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
S01-radial Offset	300	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
S01-outside Diameter	18	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
S01-projection	60	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.8; geometric allowance, not hydraulic/DN sizing.
S02-elevation	4110	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
S02-bore	15	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
S02-azimuth angle	90	deg	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.

Design feature	Saved value	Unit	Provenance / governing rule
S02-radial Offset	300	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
S02-outside Diameter	18	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
S02-projection	60	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.2; geometric allowance, not hydraulic/DN sizing.
I01-elevation	330	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I01-bore	15	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I01-azimuth angle	45	deg	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I01-radial Offset	300	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I01-outside Diameter	18	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I01-projection	60	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I02-elevation	4170	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.

Design feature	Saved value	Unit	Provenance / governing rule
I02-bore	15	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I02-azimuth angle	45	deg	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I02-radial Offset	300	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I02-outside Diameter	18	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
I02-projection	60	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bore D/40; zone fraction 0.3; geometric allowance, not hydraulic/DN sizing.
D01-elevation	0	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bottom-pole axial drain; azimuth N/A.
D01-bore	24	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bottom-pole axial drain; azimuth N/A.
D01-radial Offset	0	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bottom-pole axial drain; azimuth N/A.
D01-outside Diameter	28.8	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bottom-pole axial drain; azimuth N/A.
D01-projection	60	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bottom-pole axial drain; azimuth N/A.
D01-surface Edge Elevation Min	0	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bottom-pole axial drain; azimuth N/A.

Design feature	Saved value	Unit	Provenance / governing rule
D01-surface Edge Elevation Max	0.1729	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: bottom-pole axial drain; azimuth N/A.
V01-elevation	4710	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.
V01-bore	24	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.
V01-azimuth angle	315	deg	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.
V01-radial Offset	180	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.
V01-outside Diameter	28.8	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.
V01-projection	60	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.
V01-surface Edge Elevation Min	4704.24605	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.
V01-surface Edge Elevation Max	4715.076617	mm	Class C; System-generated ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; Class C. ECR_KUHNI_PREPILOT_GEOMETRY_RULESET_R2_ADOPTED_COMPARTMENT_EFFICIENCY; C: r=0.30D, ellipsoidal surface $z=zT+0.20D$; not absolute-high-point or vent-duty qualification.

Limitations and CAD handoff acceptance

Geometrically complete pre-pilot layout only; not fabrication or mechanical qualification.

- Class A: approved frozen Stage-3/4; no hydraulic/sizing recalculation.
- Class B: Perry 8e p15-83 / Pratt-Stevens design envelope; Garthe Table A.1 and Weber-Jupke 2020 DOI 10.1002/aic.16286 construction family.
- End zones are packaging allocations, not separation-performance sizing. Connection bores are geometric allowances, not hydraulic or DN sizing.
- NMP-down/RRBO-up is an explicit layout selection, not inferred from continuity. Vent is crown-region, not absolute-high-point qualified.
- N+1 plates follows selected shared-boundary and terminal-ring topology, not a universal source count.
- Approved component source: approved-component-manifest.json canonical SHA-256 954fd39266e55dbaa2c0c10dcc6578b5a94d9247b1ff95194f0a762b58a32042.
- PRELIMINARY INTERFACE HOLD: retain the existing vessel shell, shaft-support elevations/wall interfaces, drive/seal envelopes and nozzle arrangement pending mechanical interface review.
- Component geometry approval does not qualify support-wall attachments, stator mounting, shaft/hub attachment, strength, vibration or fabrication.
- Pressure-vessel shell/head thickness and code design
- Shaft strength, deflection, critical speed
- Bearing/seal ratings, drive power/torque and support structural qualification
- Welds, keys, fasteners, flanges, reinforcement and external pipe routing
- Separation residence-time, connection hydraulic capacity and operational venting qualification

Not specified by the approved component basis — outside scope: pressure-rated wall/head material thicknesses; materials and corrosion allowance; welding, tolerances, fits and fasteners; bearing/seal product selection; shaft strength, deflection and critical speed; drive torque/power adequacy; structural loads; nozzle flanges/reinforcement and piping qualification. Do not invent these details or issue fabrication drawings from this report.

No unresolved geometry items recorded in this saved revision. Explicit scope exclusions above are not fabrication-ready details.

PRELIMINARY PRE-PILOT GEOMETRY — NOT FOR FABRICATION